Shivansh Kalra

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EDUCATION

Shiv Nadar University, B.Tech in Computer Science — Aug 2023 - May 2027

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Rust, C++, SQL, Bash.

Backend and Infrastructure: FastAPI, Node.js, Next.js, React, PostgreSQL, Redis, Docker, AWS, GitHub Actions, SSE, WebSockets.

AI and ML: LLM APIs (Anthropic, OpenAI, Gemini), RAG, agents and evals, PyTorch, Hugging Face Transformers, FAISS, LightGBM, XGBoost, SHAP.

Systems and Security: Playwright, Chrome DevTools Protocol, network-layer replay, TOTP/HMAC, tree-sitter static analysis, lock-free concurrency, loom model checking.

EXPERIENCE

Attack Capital (YC-backed), Software Development Engineer — Remote | May 2026 - Aug 2026
HealOS, a healthcare operations platform used by 50+ US companies

- Built EHR automation on network-layer replay with Browserbase and Playwright, replaying each EHR's internal API calls instead of scraping its UI. Cut integration build time from weeks to a few hours; the approach was adopted company-wide.
- Implemented TOTP from RFC 4226/6238 (HMAC-SHA1) from scratch so automated workflows could pass authenticator prompts; also adopted company-wide.
- Led the technical expansion of HealOS into the UAE, building the integration layer for regional compliance and partner onboarding.
- Found and reported 3 critical security vulnerabilities in an EHR platform used by 40%+ of US healthcare professionals.
- Tuned ElevenLabs model configuration, chunk scheduling and streaming for production voice agents across 3 portfolio products.

Pactle, Founding Software Engineer (Contract) — Remote | Feb 2026 - Apr 2026

- Built the generalised product's frontend from its first commit and co-built the backend: lead management, SLA tracking and round-robin assignment.
- Designed a deterministic RFQ-to-quote engine (fuzzy search, multi-algorithm scoring, Tesseract OCR for scanned RFQs, approvals, multi-currency) that was integrated into Pactle's product.

Thinkly Labs, Full Stack Engineer (Internship) — Remote | Nov 2025 - Jan 2026

- Built North, a content generation engine, end to end: database schema, API design and frontend. Worked with the founding team on distributed system architecture.

Earlier: Founder, CatalystIQ (Dec 2023 - Apr 2024), B2B SaaS incl. an AI resume screener 10+ clients used on 4,000+ candidates; Software Development Intern, Frontrunner (2024); Founding Engineer, Vidya Verse (2021).

PROJECTS

Invariant | Rust, Python, Next.js, Playwright, tree-sitter — invar-v4.vercel.app

- Founder and sole engineer of a security and infrastructure analysis tool that scans an app three ways: a public-URL Playwright scan, offline analysis of a recorded login session, and a codebase call graph with 8 graph analyses. Scans its own 157-file codebase in 1.8 s; validated with 12 founders.

HippoVoice | Python, PyTorch, Qwen3-4B, Qwen2.5-Omni, ChromaDB — github.com/shivansh193/hippovoice

- Long-term memory for voice agents with decay-based salience and graph retrieval. On LoCoMo (1,540 questions) it matched a Mem0 reimplementation on recall (29.47% vs 29.11% F1) while retrieving a third as much irrelevant context.

clipy | Rust, Chrome DevTools Protocol, ChaCha20-Poly1305 — github.com/shivansh193/clipy

- Open-source CLI that records a browser session, infers data dependencies between API requests and generates a runnable replay script; reduced a 25-request flow to the 5 it needed. 33 tests, CI on Linux, macOS and Windows.

QuantEdge | Rust, SQLite, SEC EDGAR, GitHub Actions — github.com/shivansh193/quant-edge

- Multi-factor stock picker with point-in-time data, a backtester with a market-impact cost model, and a hash-chained forward-test log appended daily by CI. 194 tests.

RESEARCH AND ACHIEVEMENTS

- **Finverse 2026 Research Competition, 2nd place:** prediction markets as macro hedges across 383 Kalshi contracts; hedged Sharpe 0.538 vs 0.065.
- **Amazon ML Challenge 2026:** led a team of 3 on entity resolution (mBERT, FAISS, cross-encoder re-ranking); 0.961 macro F0.5.
- **Smart India Hackathon 2026 (NTRO):** IKEv1/v2 parser and encrypted-traffic classifier (0.79 to 0.98 macro-F1) for an IPsec analyzer.
- **Also:** forge (78 tests in 26 hours), Recourse (LexHack 2026), PACER (published a negative result on lock-free reclamation); top 3 in 5+ hackathons; BrAINWARS 2026 Top 20; published a book on blockchain at 17.